

Homework Set 1. Math Prerequisites, Dynamic Modeling, and Simulation

(Due on Friday, January 17, 2025)

1. (20 pt) Calculation with complex numbers

Message from the instructor: French mathematician Jacques Hadamard (1865–1963) once joked, “The shortest path between two truths in the real domain passes through the complex domain.” Complex numbers are necessary for understanding control. To prevent any surprise when you meet them later, let us start preparing ourselves from now, step by step, through some exercise problems in the first three homework sets. Please take this message – you will likely encounter new problems as you develop your future profession. Never be afraid or frown to learn new things, even if they are not within your “comfort zone”.

- (a) A complex number $z \in \mathbb{C}$ can be written as $z = x + yi$, where x and y are real numbers, called the *real part* and *imaginary part* of z , respectively ($x = \operatorname{Re} z$, $y = \operatorname{Im} z$). A complex number can be plotted as a point on a two-dimensional plane, called the *complex plane*. Please draw a complex plane and label the number $-1 + \sqrt{3}i$ on it.
- (b) The *modulus* (or the *absolute value*) of a complex number $z = x + yi$ is the distance between the origin and point z on the complex plane: $|z| = \sqrt{x^2 + y^2}$. The *argument* of z is the angle obtained by anticlockwise rotating the positive real axis to the vector z on the complex plane, denoted as $\angle z$. Report the modulus and argument of $-1 + \sqrt{3}i$.
- (c) Euler’s formula states that for any $\theta \in \mathbb{R}$, we have

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Please remember this formula, either by proving to yourself or looking for a proof somewhere. Based on this formula, show that $|e^{i\theta}| = 1$ always hold.

- (d) Label the number $re^{i\theta}$ ($r > 0$ and θ are both real) on the complex plane and indicate r and θ .
- (e) Testify that you have grasped the above concepts by writing down “I understand the concepts of real part, imaginary part, modulus, and argument of complex numbers, and the expression of Euler’s formula.”

2. (20 pt) Numerically solving ODE with MATLAB.

Message from the instructor: You should have learned (in CHE 225) Matlab commands `ode45` (`ode23`, `ode15s`, or similar commands in other languages) to numerically solve ordinary differential equations. Let’s recapture in case you haven’t practiced them for a while. You may also refer to Appendix C.1 of Seborg.

- (a) [Single-dimensional problem.] Solve ODE for $x = x(t)$:

$$\frac{dx}{dt} = \sqrt{x^2 + t^2}$$

from initial condition $x(0) = 0$. Report the value of x at $t = 3$. Submit your codes too.

- (b) [Multi-dimensional problem.] Simulate the Lorenz system:

$$\frac{dx}{dt} = s(y - x), \quad \frac{dy}{dt} = x(r - z) - y, \quad \frac{dz}{dt} = xy - bz$$

where $s = 10$, $r = 28$, $b = 8/3$, from initial condition $x(0) = y(0) = z(0) = 1$. Plot the trajectory of (x, y, z) in a three-dimensional curve plot. You may use MATLAB command `plot3` for this purpose.

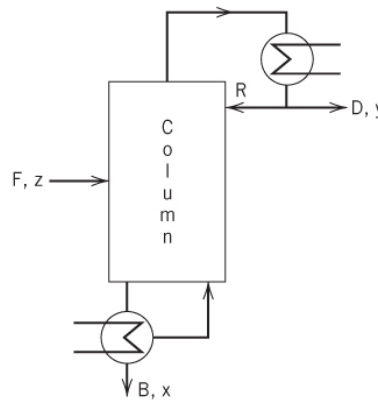
Remark: This system is a “chaotic” one that shows how complicated a nonlinear system behavior might be. See an animated result at: <https://www.youtube.com/watch?v=oqDQwEvHGfE>.

- (c) Testify that you are now able to do simulations of ODEs by writing “I understand the basic usage of Matlab commands to simulate ODEs and was able to generate the above results by my own coding.”

3. (20 pt) Feedback and feedforward control schemes. [Seborg 1.7]

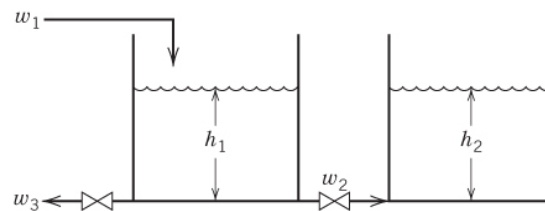
The distillation column shown in the below figure is used to distill a binary mixture. Symbols x , y , and z denote mole fractions of the more volatile component, while B , D , R , and F represent molar flow rates. It is desired to control distillate composition y despite disturbances in feed flow rate F . All flow rates can be measured and manipulated with the exception of F , which can only be measured. A composition analyzer provides measurements of y .

- (a) Propose a feedback control method and sketch the schematic diagram.
 (b) Suggest a feedforward control method and sketch the schematic diagram.



4. (15 pt) Tank system modeling. [Seborg 2.3]

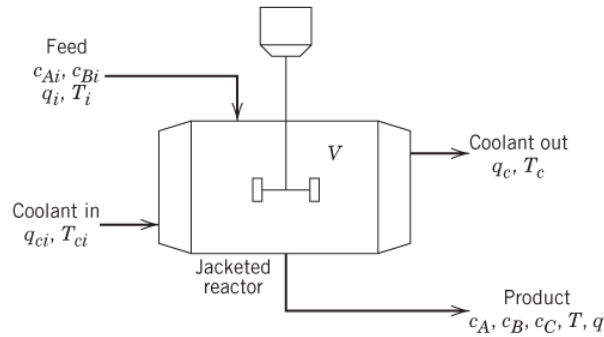
Two tanks are connected together in the following unusual way in the figure below.



- The density of the incoming liquid, ρ , is constant.
 - The cross-sectional areas of the two tanks are A_1 and A_2 .
 - w_2 is positive for flow from Tank 1 to Tank 2.
 - The two valves are linear with resistances R_2 and R_3 .
- (a) Develop a model for this system that can be used to find h_1 , h_2 , w_2 , and w_3 as functions of time for any given variations in inputs.
 (b) Perform a degrees of freedom analysis. Identify all input and output variables.

5. (25 pt) Reactor modeling. [Seborg 2.9]

Irreversible consecutive reactions $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ occur in a jacketed, stirred-tank reactor, as shown in the figure below.



Derive a dynamic model based on the following assumptions:

- The contents of the tank and cooling jacket are well mixed. The volumes of material in the jacket and in the tank do not vary with time.
- The reaction rates are given by

$$r_1 = k_1 \exp(-E_1/RT) c_A \text{ [mol A/(h}\cdot\text{L)]}, \quad r_2 = k_2 \exp(-E_2/RT) c_B \text{ [mol B/(h}\cdot\text{L)]}$$
- The thermal capacitances of the tank contents and the jacket contents are significant relative to the thermal capacitances of the jacket and tank walls, which can be neglected.
- Constant physical properties and heat transfer coefficients can be assumed.

Note: All flow rates are volumetric flow rates in L/h. The concentrations have units of mol/L. The heats of reaction are ΔH_1 and ΔH_2 .

Message from the instructor: In each homework set, there are 2 bonus problems worth 20 bonus points relative to the 100 points of the main problems. The first bonus problem, labeled as “**process control in practice**”, aims to give you some basic domain knowledge about how control works in real-world applications. Some textbook chapters cover these topics, but we don’t cover all these in the lectures, because (i) not everyone is going to be a control engineer, and (ii) we should focus on the principles, not those petty details (that you will be trained on when you really have a job on it). If you have an interest, you are encouraged to read these chapters and do the problem. The second bonus problem, labeled as “**process control in depth**”, poses some more extensive problems that challenge your skills on coding, mathematical derivation, and theoretical reasoning. The aim is to give you a better idea of the scientific depth of control theory, and encourage you to develop computational and logical skills and understand things more profoundly.

6. (Bonus: 10 pt) Process control in practice: Control system instrumentation.

List the main categories of pressure-measuring devices and explain the differences among their mechanisms. How are different types of pressure sensors suitable for different applications? *You may want to read Section 9.1 of Seborg and/or look for resources on the Internet. You should summarize your points, rather than copying a large volume of text.*

7. (Bonus: 10 pt) Process control in depth: Computer simulation of a dynamic response.

Read Example 2.5 of Seborg, and *reproduce the simulation results in Figure 2.7*. Describe qualitatively the nonlinear behavior of the CSTR that you observed in the simulations. *Submit your codes.*